

1. SIGNAL GENERATION AND BASIC SIGNAL OPERATIONS

Experiment 1.1

Generation of Continuous Time Signals

Program:

```
% Generation of Continuous Time Signals
```

```
clc;
```

```
clear;
```

```
close all;
```

```
% Time axis
```

```
t = -5:0.01:5;
```

```
% Continuous-time sine signal
```

```
x = sin(2*pi*t);
```

```
% Plotting
```

```
figure;
```

```
plot(t, x, 'LineWidth', 2);
```

```
grid on;
```

```
title('Continuous Time Sine Signal');
```

```
xlabel('Time (seconds)');
```

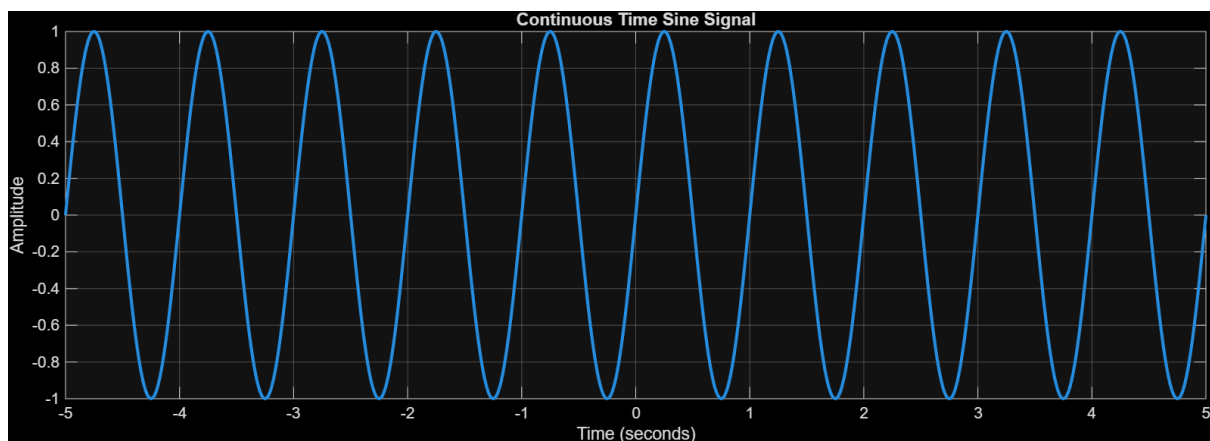
```
ylabel('Amplitude');
```

Expected Output:

A continuous-time sine wave is displayed with:

- Time range: -5 s to 5 s
- Amplitude range: -1 to 1
- Periodic sinusoidal waveform

Output Waveform:



Result:

This test case validates that the signal generator can produce a sinusoidal waveform with

Experiment 1.2

Generation of Discrete Time Signals

Program:

```
% Generation of Discrete Time Signal
```

```
clc;
```

```
clear;
```

```
close all;
```

```
% Discrete-time index
```

```
n = -10:10;
```

```
% Discrete-time sinusoidal signal
```

```
x = sin(0.2*pi*n);
```

```
% Plotting
```

```
figure;
```

```
stem(n, x, 'filled');
```

```
grid on;
```

```
title('Discrete Time Sinusoidal Signal');
```

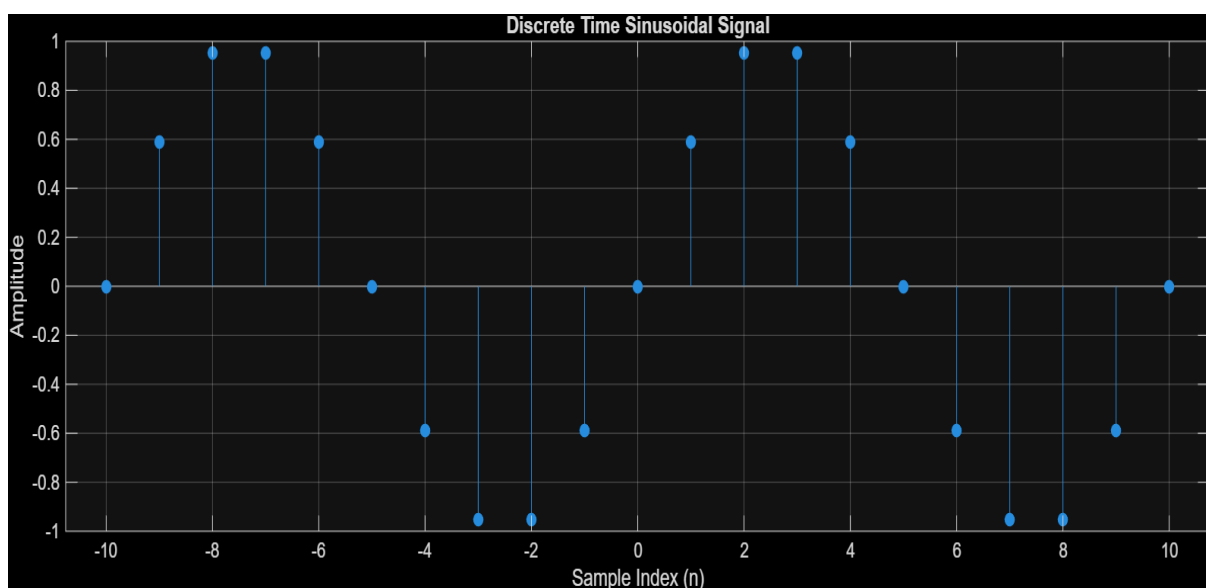
```
xlabel('Sample Index (n)');
```

```
ylabel('Amplitude');
```

Expected Output:

A discrete-time sinusoidal signal is displayed using stem representation, showing individual samples at discrete time instants.

Output Waveform:



Result:

This test case validates that the signal generator can accurately produce a discrete-time unit impulse signal at a specified index, which is essential for analyzing the response of discrete time systems.

Experiment 1.3

Generation of Triangular and Sawtooth Signals

Program :

% Generation of Triangular and Sawtooth Signals

clc;

clear;

close all;

% Time axis

t = 0:0.01:2;

% Sawtooth Signal

x1 = sawtooth(2*pi*t);

% Triangular Signal

x2 = sawtooth(2*pi*t,0.5);

% Plotting

figure;

subplot(2,1,1);

plot(t,x1,'LineWidth',2);

grid on;

title('Sawtooth Signal');

xlabel('Time (s)');

ylabel('Amplitude');

subplot(2,1,2);

plot(t,x2,'LineWidth',2);

grid on;

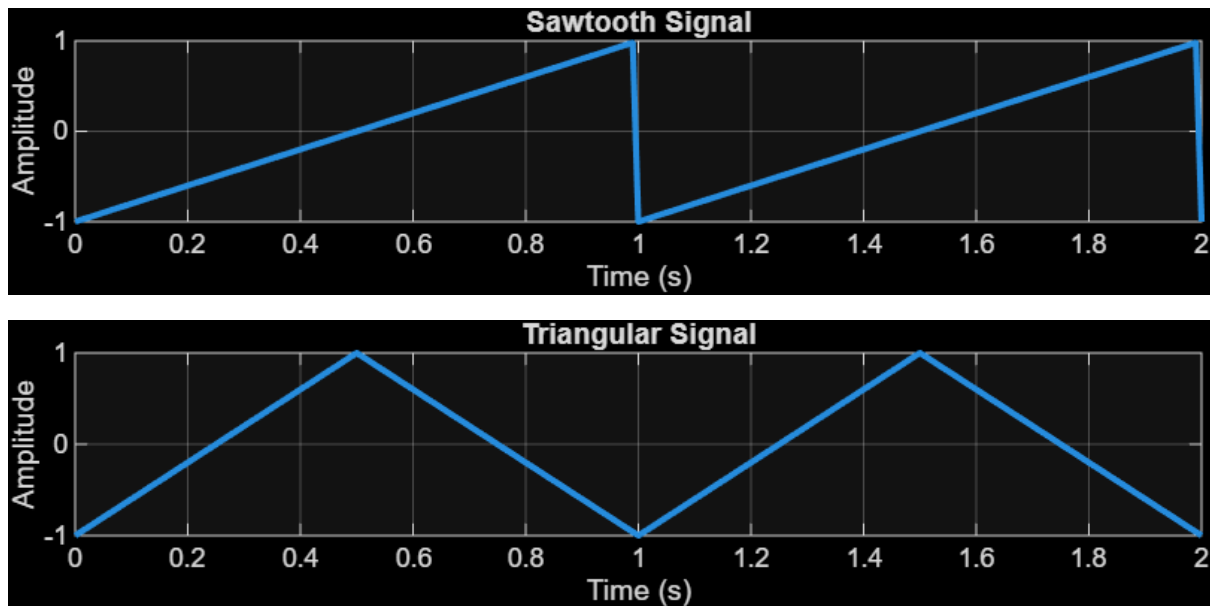
title('Triangular Signal');

xlabel('Time (s)');

ylabel('Amplitude');

Output Generated Graph :

Two plots: symmetric triangular waveform and ramp-like sawtooth waveform.



Result:

Triangular wave has equal rise and fall; sawtooth shows sharp drop and gradual rise.

Experiment 1.4

Step and Ramp Functions (Continuous & Discrete) .

Program:

```
% Step and Ramp Functions (Continuous & Discrete) clc;
clear;
close all;
%% Continuous-Time
Signals t = -5:0.01:5;
% Continuous Unit Step
Function u = double(t >= 0);
% Continuous Ramp
Function r = t .* (t >= 0);
%% Discrete-Time Signals
n = -5:5;
% Discrete Unit Step
Function ud = double(n >= 0);
```

```

% Discrete Ramp Function
rd = n .* (n >= 0);
%% Plotting
figure;
subplot(2,2,1);
plot(t,u,'LineWidth',2);
grid on;
title('Continuous Unit Step
Function'); xlabel('Time (t)');
ylabel('Amplitude');
subplot(2,2,2);
plot(t,r,'LineWidth',2);
grid on;
title('Continuous Ramp
Function'); xlabel('Time (t)');
ylabel('Amplitude');
subplot(2,2,3);
stem(n,ud,'filled');
grid on;
title('Discrete Unit Step
Function'); xlabel('Sample Index
(n)');
ylabel('Amplitude');
subplot(2,2,4);
stem(n,rd,'filled');
grid on;
title('Discrete Ramp Function');
xlabel('Sample Index (n)');
ylabel('Amplitude');

```

Output Generated Graph:

